Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 4 [2023 P2]

Test Date: 16th Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2023-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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1. The volume of the parallelopiped whose coterminous edges are represented by the vectors $2\vec{b}\times\vec{c}, 3\vec{c}\times\vec{a}$ and $4\vec{a}\times\vec{b}$ where $\vec{a}=(1+\sin\theta)\hat{i}+\cos\theta\hat{j}+\sin 2\theta\hat{k}$, $\vec{b}=\sin\left(\theta+\frac{2\pi}{3}\right)\hat{i}+\cos\left(\theta+\frac{2\pi}{3}\right)\hat{j}+\sin\left(2\theta+\frac{4\pi}{3}\right)\hat{k}, \vec{c}=\sin\left(\theta-\frac{2\pi}{3}\right)\hat{i}+\cos\left(\theta-\frac{2\pi}{3}\right)\hat{j}+\sin\left(2\theta-\frac{4\pi}{3}\right)\hat{k}$ is 18 cubic units, then which of the following value in the interval $\left(0,\frac{\pi}{2}\right)$ is not possible for θ:

A) $\frac{\pi}{9}$ **B)** $\frac{2\pi}{9}$ **C)** $\frac{\pi}{3}$ **D)** $\frac{4\pi}{9}$

2. If Z_1, Z_2, Z_3 are all unimodular also if $Z_1.\left(\overline{Z}_2+\overline{Z}_3\right)=$ purely imaginary, then possible value of $\left|Z_1+Z_2+Z_3\right|$ will be:

A) 1 **B)** 2 **C)** $\sqrt{3}$ **D)** 0

3. Let $A=\left[a_{ij}\right]_{n\times n}$ be a matrix such that $a_{ij}=i.2^{-j}$, then the value of $\lim\limits_{n\to\infty}\left(trace\,A^n\right)^{1/n}$ (where $n\in N$) is equal to

A) 2 **B)** 1.5 **C)** 1 **D)** e

4. Let $\left(1+x+x^2\right)^{20}=a_0+a_1x+a_2x^2+.....+a_{40}x^{40}$. Further $S=\sum\limits_{r=0}^{40}a_r$. Two coefficients are chosen from the coefficients $a_0, a_1, ..., a_{40}$ and the probability that they are equal is p (considering no three coefficients are equal). Units digit of S is equal to b and $a=\frac{1-p}{10p}$. On the basis of above information, answer the following:

If $\left(\sqrt{a-b}+b\right)^6=I+F$, where $0\le F<1$ and $I\in N$, then the value of I is

A) 413 **B)** 414 **C)** 415 **D)** 416

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated <u>according to the following marking scheme :</u>

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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5. If the equation $x^2+(2-\tan\theta)x-(1+\tan\theta)=0$ has two integral roots then possible sum of any two distinct values of $\theta\in[0,2\pi]$ will be:

A) π **B)** 2π **C)** $\frac{3\pi}{2}$ **D)** $\frac{5\pi}{2}$

6. P is a point on $y^2=12x$, such that its focal distance is 6. Tangent drawn at P intersect tangent at vertex at point T(y > 0). Point R is on x-axis inside parabola such that SR = 6 (S is focus). M is a point on parabola such that tangents at M & P meet at point N such that SN = 5. then :

A) Area of quadrilateral PRST is 18 units.

B) Area of quadrilateral PRST is 27 units.

C) Sum of squares of possible ordinates of M is 28.

D) Sum of squares of possible ordinates of M is 30.

7. A triangle ABC has sides AB = 13, BC = 5 and AC = 12 units, slides on XY axes such that A & B remains on positive X and positive Y-axis respectively. then :

A) Locus of C is $x^2+y^2=169$

B) Locus of C is 12x = 5y

C) If p. OA + q .OB = OC, then $\left[\frac{1}{p}+\frac{1}{q}\right]=3$

D) If p. OA + q .OB = OC, then $\left[\frac{1}{p}+\frac{1}{q}\right]=2$ where [.] is G.I.F.)

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases.*

8. If the possible solution of the differential equation $y\left(y^2-2x^2\right).dx+x.\left(2y^2-x^2\right).dy=0$ is of the form f (x). g (y). (g (y) – f (x)) = C (C is arbitrary constant) where f (x), g (y) are monic polynomials. Then find value of $\frac{4.g(3)}{f(2)}$

9. If planes : $P_1 \equiv x = y\sin\alpha + z\sin\beta$

$P_2 \equiv y = x\sin\alpha + z\sin\theta$

$P_3 \equiv z = x\sin\beta + y\sin\theta$

pass through the line $\frac{x}{\cos\theta}=\frac{y}{\cos\beta}=\frac{z}{\cos\alpha}$, then sum of possible values of $\alpha+\beta+\theta$ is K. (where $\alpha+\beta+\theta\in[0,6\pi]$)

Also $\sin^2\theta+\sin^2\alpha+\sin^2\beta=p+q.\sin\alpha.\sin\beta.\sin\theta$, then find $p+q+[K]$.

([.] represents greatest integer function)

10. If $A=\left[a_{ij}\right]_{2\times2}, B=\left[b_{ij}\right]_{2\times2}$ such that $a_{ij}=f\left(x^{2i-j-2}\right)$ and $b_{ij}=f\left(x^{2i-j-1}\right)$ and

$$A+B^T=\begin{pmatrix} -x^2 f\left(x+f(1)\right) & \left(1-\frac{1}{x^4}\right)f\left(\frac{1}{x^2}\right) \\ f\left(\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right) & -\frac{1}{x^2}f\left(\frac{1+f(1)}{x}\right)\end{pmatrix}$$, then $\int_x^{1/x} f(t)\,dt$ is equal to_____

11. There are 10 girls and 8 boys in a class room including Mr. Ravi, Ms. Rani and Ms. Radha. A list of speakers consisting of 8 girls and 6 boys has to be prepared. Mr. Ravi refuses to speak if Ms. Rani is a speaker. Ms. Rani refuses to speak if Ms. Radha is a speaker. The number of ways the speakers can be chosen is a 3 digit number $n_1n_2n_3$, then $|n_3+n_2-n_1|=$

12. If $f(x)=\begin{cases} e^x+e^{-x}, & x<0 \\ \log_{10}(100-x), & x\in[0,100)\end{cases}$ then the number of solution of the equation

$f(x)+\left(e-\frac{1}{e}\right)x=\frac{2}{e}$

13. If the maximum value of $\left|\sqrt{x^4-7x^2-4x+20}-\sqrt{x^4+9x^2+16}\right|$ is P, find the value of [P] (where [.] denotes greatest integer function)

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
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- **Zero Marks:0** in all other cases

PARAGRAPH-I

$$f(x)=[x-1].x^2+\left(a^2+ab+2\right).\text{sgn}(x-2)+(a+b).\{x\}$$

([.] denotes GIF and {.} Fractional part function)

14. If f (x) is continuous at x = 2 then b equals_____.

15. If f (x) has isolated point discontinuity at x = 2 then smallest positive integral value of b is, provided $a \in R$.

PARAGRAPH-II

If f (x) is continuous & differentiable function, $f(x) \neq 0$ for all x in interval [4, 8] given

$$f(4)=\frac{1}{4}, f(8)=\frac{1}{2} \text{ and } \int_4^8 \frac{\left(f^1(x)\right)^2}{\left(f(x)\right)^4}.\,dx=1$$

16. Find value of $\dfrac{1}{5.f(6)}$

17. Find value of $\dfrac{\int_0^1 \left(\prod_{r=4}^{11} \dfrac{2}{f(-x+r)}\right) \times \left(\sum_{r=4}^{11} f(-x+r)\right).dx}{8!}$

PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

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- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
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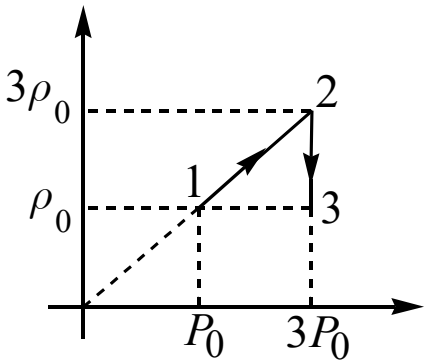
Full Marks : +3 If **ONLY** the correct option is chosen;
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18. The length of a wire required to manufacture a solenoid of length 'ℓ' and self induction 'L' is found to be $\sqrt{p}\sqrt{\frac{\pi L\ell}{\mu_0}}$, (Cross sectional area of the wire is negligibly small). Find the value of 'p'

A) 4 **B)** 2 **C)** 8 **D)** 0.5

19. The density ρ v/s pressure P graph of an ideal monoatomic gas undergoes a process as shown in the figure. Find total work done during the complete process. Molecular weight of gas is M and number of moles is 1.



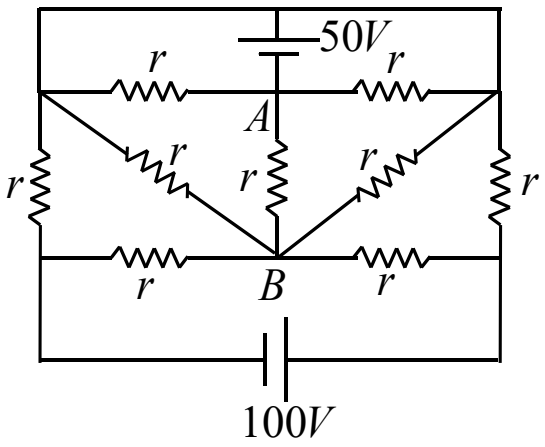
A) $\frac{P_0M}{\rho_0}(1-\ell n3)$ **B)** $\frac{P_0M}{\rho_0}(2-\ell n3)$ **C)** $\frac{P_0M}{\rho_0}(3-\ell n3)$ **D)** $\frac{P_0M}{\rho_0}(4-\ell n3)$

20. A plane mirror in xz plane is moving with a velocity $-2\hat{i}+3\hat{j}+\hat{k}$ m/s. Velocity of image of a point object moving with velocity $\hat{i}+2\hat{j}-5\hat{k}$ m/s is (assume that object is located on front side of the mirror)

A) $-2\hat{i}+4\hat{j}-\hat{k}$ m/s **B)** $\hat{i}+4\hat{j}-5\hat{k}$ m/s

C) $\hat{i}-4\hat{j}+5\hat{k}$ m/s **D)** $-\hat{i}+5\hat{j}-4\hat{k}$ m/s

21. In the given circuit, each resistance is $r=18.75\Omega$. The current (in A) in the resistance connected across A and B is '……' A.



A) 3 **B)** 2 **C)** 5 **D)** 0.6

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

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22. A moving positively charged particle is spotted at point (5 cm, 0, 0) at a given instant. At the same instant, magnetic field produced by it at origin is detected to be $\overline{B} = B_0 \hat{j}$; where B_0 is a positive constant. The possible velocities of the charged particle are

A) $\bar{v} = 5\hat{i} + 4\hat{j}\ m/s$ **B)** $\bar{v} = 2\hat{i} - 4\hat{k}\ m/s$

C) $\bar{v} = -\hat{i} - 3\hat{k}\ m/s$ **D)** $\bar{v} = -2\hat{j} - 2\hat{k}\ m/s$

23. A proton of charge 'e' and mass 'm' enters a uniform and constant magnetic field $\vec{B} = B\hat{i}$ with an initial velocity $\vec{V} = V_{ox}\,\hat{i} + V_{oy}\,\hat{j}$. Which of the following will be **CORRECT** at any later time 't'?

A) x–component of velocity of the proton will be $V_{ox}\cos\left(\frac{eBt}{m}\right)$

B) y–component of velocity will be $V_{oy}\cos\left(\frac{eBt}{m}\right)$

C) z–component of velocity will be $V_{oy}\sin\left(\frac{eBt}{m}\right)$ along $\left(+\hat{k}\right)$ direction

D) x–component of velocity will remains unchanged

24. A circular beam of charge of radius a consists of electrons moving with a constant speed u along the $+z$ direction. The beam's axis is coincident with the z–axis and the electron charge density is given by $\rho = cr^2\left(C.m^{-3}\right)$ where c is a constant and r is the radial distance from the axis of the beam: (Neglect the interaction between electrons)

A) The current crossing the xy–plane is $\pi c u a^4$

B) The electric field at distance $r\left(r < a\right)$ from z–axis is proportional to r^3

C) The magnetic field at distance $r\left(r > a\right)$ from z–axis is proportional to r^3

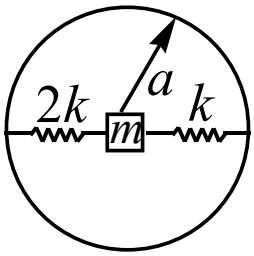
D) The electric field at distance $r\left(r < a\right)$ from z–axis is inversely proportional to r

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
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Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases..*

25. A ring of radius $a = 1m$ and having mass $4kg$ lies on a smooth horizontal surface. One end of each of the two massless spring k and $2k\left(k = 240\,N/m\right)$ are fixed to the diametrically opposite points of ring and the other end of the two springs is attached to a block of mass $m = 1kg$, which lies at the centre of the ring and in this position springs are at their natural length. If the block is displaced by a distance $a/2$ along the length of springs such that spring of spring constant k is compressed $a/2$ and is released then at the instant when each spring achieves it's natural length, the velocity of the ring $\left(in\ m/s\right)$ will be



26. Two coherent point sources S_1 and S_2 emit light of wavelength $\lambda = 400\,nm$ in same phase. The separation between the sources is 3λ. Consider a line passing through S_2 and perpendicular to the line S_1S_2. If the smallest distance from S_2 where a minimum intensity is observed is $k \times 10^{-7}m$, what is the value of k to the nearest integer?

27. A large tank of cross–section area A contains liquid of density ρ. A cylinder of density $\rho/4$, length $\ell = 40\,cm$, and cross–section area $a\left(<< A\right)$ is floating in the liquid. It is

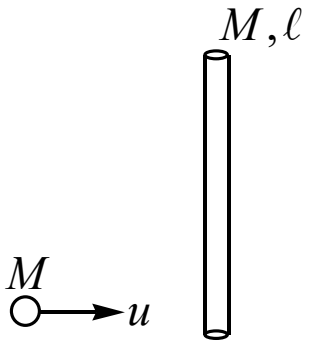
pushed inside the liquid so that it is just submerged and then released at $t = 0$. Assuming that the water level in the tank remains constant, determine the speed $(in\ m/s)$ of the cylinder when it reaches its equilibrium position for the first time. Neglect viscous forces.

28. A vertical cylinder of cross–sectional area $S = 10^{-3} m^2$ contains 1 mol of an ideal monoatomic gas under a piston of mass M = 10 kg. At a certain instant, a heater is switched on which transmits 3500J of heat per second to the gas. If the entire system is thermally insulated and the pressure of the gas under the piston remains constant, then what is the speed (in m/s) of the piston? Take atmospheric pressure to be $P_0 = 10^5 Pa$.

29. A metal block is placed in a room which is at $10^\circ C$ from long time. Now it is heated by an electric heater of power 500 W till its temperature becomes $50^\circ C$. Its initial rate of rise of temperature is $2.5^\circ C/s$. The heated is switched off and now a heater of 100W is required to maintain the temperature of the block at $50^\circ C$. The heat radiated per second when the block was at $30^\circ C$ is given as α watt. Find the value of $\left(\frac{\alpha}{10}\right)$.

(Assume Newton's law of cooling to be valid)

30. A rod of mass M and length ℓ is at rest on plane horizontal smooth surface. A particle of same mass M strike one end with velocity u perpendicular to rod, elastically. Now just after collision if the kinetic energy of upper half part of rod is $\frac{\alpha M u^2}{100}$ then find the value of α.

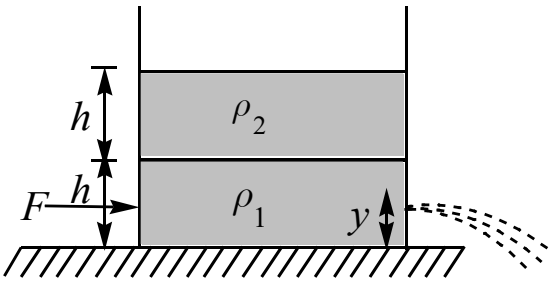


SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
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PARAGRAPH-I

A cylindrical tank having a cross–sectional area $A = 0.5m^2$ is filled with two liquids of density $\rho_1 = 900kgm^{-3}$ and $\rho_2 = 600kgm^{-3}$ to a height $h = 60cm$ each as shown in the figure. A small hole having an area $a = 5cm^2$ is made in right vertical wall at a height $y = 20cm$ from the bottom. A horizontal force F is applied to the tank to keep it in static equilibrium. The tank is lying on a horizontal surface. Neglect the mass of the cylindrical tank in comparison to the mass of the liquids (take $g = 10ms^{-2}$, Neglect flow velocity inside the tank)

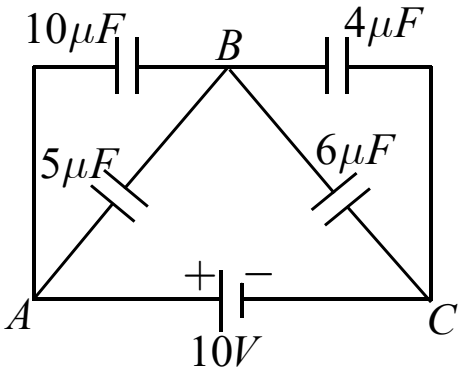


31. The velocity of efflux is $(in\ m/s)$

32. Horizontal force F to keep the cylinder in static equilibrium, if it is placed on smooth horizontal plane, is $(in\ N)$

PARAGRAPH-II

In the circuit shown in figure, four capacitors are connected to a battery.



33. The equivalent capacitance across terminals of cell is $(in\ \mu F)$

34. The charge on the $5\mu F$ capacitor is $(in\ \mu C)$

CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
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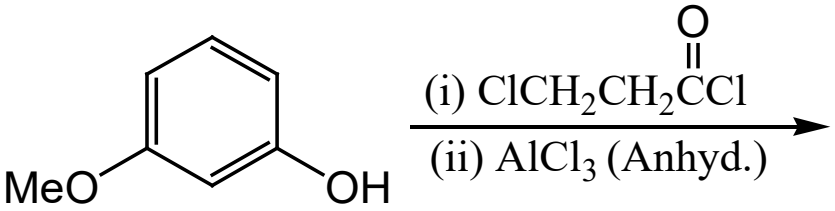
35. The wave number of electromagnetic radiation emitted during the transition of electron in between two levels of Li^{2+} ion having sum of the principal quantum numbers 4 and difference is 2, is (R_H = Rydberg constant)

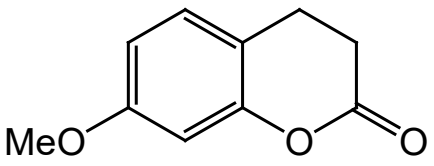
A) 3.5 R_H **B)** 4 R_H **C)** 8 R_H **D)** (8/9) R_H

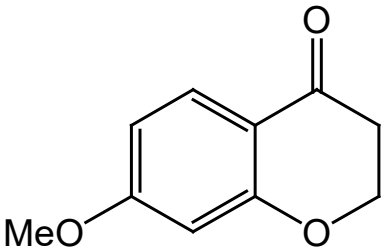
36. The number of hydrogen bonds present in the sequence of a stretch of a double helical DNA with one strand as 5^1 ATGCCTAA 3^1 is

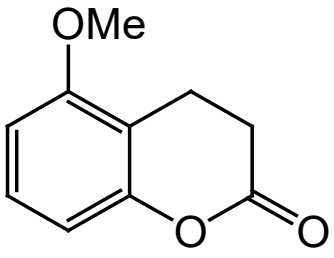
A) 16 **B)** 19 **C)** 20 **D)** 17

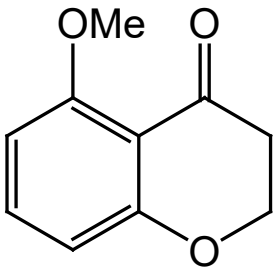
37. The major product of the following reaction is:



A)


B)


C)


D)


38. Which of the following statements BEST describes the reaction between sulfur tetrachloride $\left(SCl_4\right)$ and silver fluoride (AgF) in a suitable organic solvent?

A) The reaction proceeds via a free radical mechanism, yielding primarily SF_3Cl and AgCl.

B) The reaction is a complex equilibrium, producing a mixture of SF_4, SCl_3F, SCl_3F_2, $SClF_3$, and AgCl, with the product distribution independent on stoichiometry and reaction conditions.

C) The reaction exclusively yields SF_4 and AgCl, regardless of the stoichiometry of the reactants.

D) The driving force for the reaction is the formation of the more stable S-F bonds compared to the S-Cl bonds in SCl_4, coupled with the precipitation of AgCl, which removes a product from the equilibrium and favors product formation.

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated <u>according to the following marking scheme :</u>

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. In hydrogen−oxygen fuel cell, the following reactions take place:

Anode reaction: $2H_2\left(g\right)+4OH^-\left(aq\right)\rightarrow 4H_2O\left(l\right)+4e^-$

Cathode reaction: $O_2\left(g\right)+2H_2O\left(l\right)+4e^-\rightarrow 4OH^-\left(aq\right)$

Overall reaction: $2H_2\left(g\right)+O_2\left(g\right)\rightarrow 2H_2O\left(l\right)$

The overall reaction has a value of $\Delta H^\circ=-285.8kJ$ and $\Delta G^\circ=-237.39kJ$ at $25^\circ C$ per mole of $H_2O\left(l\right)$.

Select the correct statements

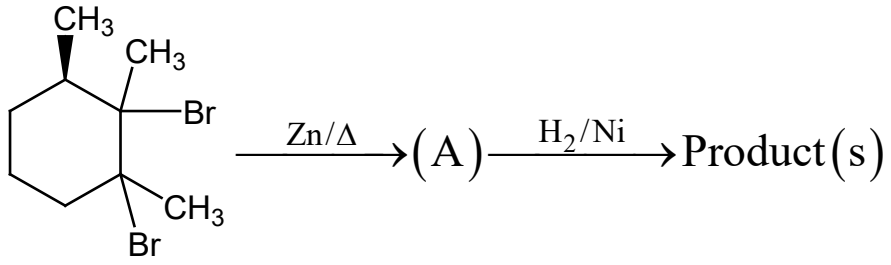
A) The standard EMF of the cell is 1.23V

B) When the concentration of hydroxide ion in the cell is doubled, cell volatage will be halved

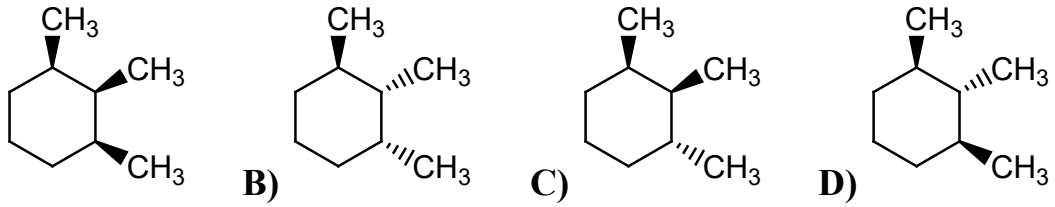
C) Theoretical efficiency of the fuel cell is given by 83.06%

D) When the concentration of hydroxide ion in the cell is doubled, cell volatage will be quadrupled

40.



Identify the products



41. A metal salt gives a colourless gas (G) with dilute HCl. The gas (G) is also obtained by treating sodium sulphite with zinc in alkali. The aqueous solution of metal salt can produce a precipitate with:

A) $Pb(CH_3COO)_2$ **B)** $BaCl_2$

C) Alkaline $Na_2[Fe(CN)_5NO]$ **D)** $AgNO_3$

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases.*

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42. The desorption of gas molecules from the adsorbent surface obeys Arrhenius equation. The average time (in sec) up to which a N_2 molecule may remain adsorbed at Pt- surface at 400 K is (Arrhenius parameters for desorption are:

$A = 1.25 \times 10^8 s^{-1}, E_a = 16 k\,cal / mol; e^{20} = 5 \times 10^8$)

43. Two elements A and B having molecular formula AB_2 and AB_4. When dissolved in 20 g of C_6H_6, 1 g of AB_2 lowers the freezing point by 2.55 K, whereas 1.0 g of AB_4 lowers it by 1.7 K. The molar depression constant for benzene is 5.1 K-kg mol^{-1}. What is the ratio of atomic mass of A to B? Assume no dissociation or association of compounds in benzene

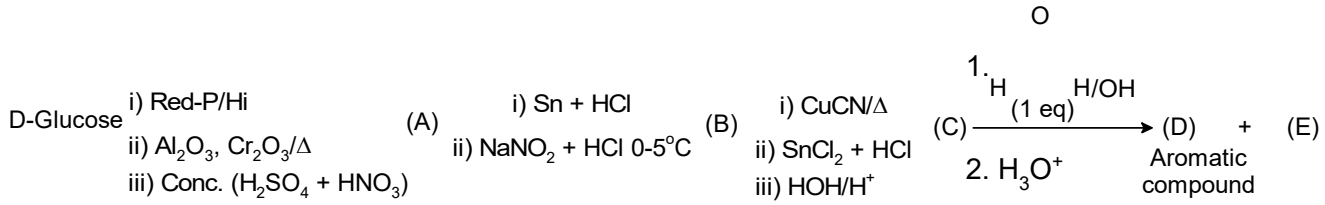
44. An ideal ionic crystal is having rock salt structure $\left(A^{+}B^{-}\right)$ such that A^{+} is occupying octahedral void. The crystal is doped with atoms of C which occupy all the tetrahedral voids without distorting the lattice. If the crystal shows defect such that the body cantered atom is missing, then calculate fraction of body diagonal covered by species in the defective crystal. Give your answer by multiplying the fraction by 10 and rounding off to nearest integer.

[Use: $\sqrt{1.5}=1.2$]

45. The decomposition reaction $3A(g) \rightarrow 2B(g) + 2C(s)$ follows first-order kinetics. Starting with pure 'A' (at 6 atm), the pressures of system after 20 minutes and after a very long time are 5.05 atm and 4.05 atm, respectively. What is the time (in min) for 93.75% A to consumed in the reaction?

46. Iron (III) chloride on reaction with Ammonium thiocyanate forms a complex(P). (P) on adding excess NaF forms colourless complex (Q). The spin only magnetic moments of complexes (P) and (Q) are 'a' BM and 'b' BM respectively. What is thevalue of (a + b) to a nearest integer?

47.



What is the molecular mass (g/mol) of compound D?

Atomic mass C=12, H=1, O =16

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

A chemist is tasked with synthesizing a copolymer composed of styrene and isoprene. They want the final product to have a specific composition: 60% styrene and 40% isoprene by weight. Furthermore, the chemist aims to control the molecular weight of the copolymer by adding a chain transfer agent, carbon tetrachloride $\left(CCl_4\right)$. They know that

for every mole of CCl_4 added, one polymer chain is terminated. Finally, they produced 10 Kg of copolymer of average molecular mass 50000u. Given molecular weight of styrene is 104 g/mol and isoprene is 68 g/mol

48. What should be the value of moles of styrene/ moles of isoprene in the initial monomer feed?

49. If moles of CCl_4 should they add per mole of total monomers in the initial feed is $x \times 10^{-2}$. Then what is the value of x is ?

PARAGRAPH-II

Substance (*A*) is a gas with vapour density 8.5. On oxidation at high pressure andtemperature with a platinum catalyst, it gave a colourless gas (*B*), which rapidly turned brown in airforming a gas (*C*). *B* and *C* were condensed together to give substance (*D*), which reacted with waterforming a compound (*E*). *E*turned blue litmus paper red. On treatment of *E* with acidified solution of KI, gas *B* had evolved, but when *E* was treated with a solution of NH_4Cl (at suitable temperature), a stable, colourless gas (*F*) had evolved. *F* did not support combustion, but magnesium continued to burn in it. However, *F* reacted with calcium-carbide in an electric furnace, forming a solid (*G*), which was slowly hydrolysed by water, forming a solution of substance *A*.

Where necessary use, Atomic mass O= 16, P=31, S =32, H=1, C =12, Ca=40,

50. Maximum number of atoms that can be on a plane in any of the conformations of a molecule of D is ___

51. How many moles of (F) is required to react with 64 grams of Calcium carbide to form G with 100% efficiency?